

Lagrangian Function and Duality

In optimization, we often want to minimize a function subject to constraints. The *Lagrangian method* allows us to combine the objective and constraints into a single function. From it, we can define the *dual problem*, which provides lower bounds to the original problem and, under convexity, has the same solution (strong duality).

2. The Lagrangian Function

For a constrained problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0 \quad (i = 1, \dots, m), \quad h_j(x) = 0 \quad (j = 1, \dots, p),$$

the Lagrangian is

$$L(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x),$$

where $\lambda_i \geq 0$ are multipliers for inequalities and ν_j are free multipliers for equalities.

The *dual function* is

$$q(\lambda, \nu) = \inf_x L(x, \lambda, \nu),$$

and the *dual problem* is

$$\max_{\lambda \geq 0, \nu \text{ free}} q(\lambda, \nu).$$

1 Linear Programming (LP)

Canonical Primal–Dual Pairs (minimization form)

(A) Constraints of type $\geq$ with $x \geq 0$

$$\begin{array}{ll} \text{Primal (P):} & \min_{x \geq 0} c^\top x \\ & \text{s.t. } Ax \geq b \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \text{Dual (D):} & \max_{y \geq 0} b^\top y \\ & \text{s.t. } A^\top y \leq c \end{array}$$

(B) Equality constraints with $x \geq 0$

$$\begin{array}{ll} \text{Primal:} & \min_{x \geq 0} c^\top x \\ & \text{s.t. } Ax = b \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \text{Dual:} & \max_{y \text{ free}} b^\top y \\ & \text{s.t. } A^\top y \leq c \end{array}$$

(C) Constraints of type $\leq$ with $x \geq 0$

$$\begin{array}{ll} \text{Primal:} & \min_{x \geq 0} c^\top x \\ & \text{s.t. } Ax \leq b \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \text{Dual:} & \max_{y \leq 0} b^\top y \\ & \text{s.t. } A^\top y \leq c \end{array}$$

Sign cheatsheet (min-form LP). Constraint $\geq \Rightarrow y \geq 0$; $\leq \Rightarrow y \leq 0$; $= \Rightarrow y$ free. Variable $x \geq 0 \Rightarrow$ dual constraint “ $\leq$ ”; $x \leq 0 \Rightarrow$ dual constraint “ $\geq$ ”; x free $\Rightarrow$ dual constraint “ $=$ ”.

2 Convex Quadratic Programming (QP) with $Q \succ 0$

(A) Inequality constraints only

$$\begin{aligned} \text{Primal : } \quad & \min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top Q x + c^\top x \\ \text{s.t. } \quad & A x \geq b. \end{aligned}$$

Lagrangian: $L(x, \lambda) = \frac{1}{2} x^\top Q x + c^\top x + \lambda^\top (b - A x)$, $\lambda \geq 0$. Stationarity: $Q x + c - A^\top \lambda = 0 \Rightarrow x^* = Q^{-1}(A^\top \lambda - c)$. Dual function:

$$q(\lambda) = -\frac{1}{2} (A^\top \lambda - c)^\top Q^{-1} (A^\top \lambda - c) + b^\top \lambda.$$

$$\text{Dual: } \max_{\lambda \geq 0} -\frac{1}{2} (A^\top \lambda - c)^\top Q^{-1} (A^\top \lambda - c) + b^\top \lambda.$$

(B) Mixed equalities and inequalities

$$\begin{aligned} \text{Primal : } \quad & \min_x \frac{1}{2} x^\top Q x + c^\top x \\ \text{s.t. } \quad & A x \geq b, \quad E x = d. \end{aligned}$$

Lagrangian: $L(x, \lambda, \nu) = \frac{1}{2} x^\top Q x + c^\top x + \lambda^\top (b - A x) + \nu^\top (d - E x)$, with $\lambda \geq 0$ (for inequalities), $\nu \in \mathbb{R}^p$ (free for equalities). Stationarity: $Q x + c - A^\top \lambda - E^\top \nu = 0 \Rightarrow x^* = Q^{-1}(A^\top \lambda + E^\top \nu - c)$. Dual function:

$$q(\lambda, \nu) = -\frac{1}{2} (A^\top \lambda + E^\top \nu - c)^\top Q^{-1} (A^\top \lambda + E^\top \nu - c) + b^\top \lambda + d^\top \nu.$$

$$\text{Dual: } \max_{\lambda \geq 0, \nu \text{ free}} -\frac{1}{2} (A^\top \lambda + E^\top \nu - c)^\top Q^{-1} (A^\top \lambda + E^\top \nu - c) + b^\top \lambda + d^\top \nu.$$

Notes. $Q \succ 0$ ensures a unique inner minimizer and a concave quadratic dual. Add simple bounds (e.g., $x \geq 0$) by appending rows to $A x \geq b$ (e.g., $-I x \geq 0$). Under Slater's condition, strong duality holds.

Example A:

$$\min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

Constraint as $g(x) = 1 - x \leq 0$.

$$L(x, \lambda) = x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

For fixed λ : minimize $x^2 - \lambda x + \lambda$. Stationary point: $x = \lambda/2$. Plug in:

$$q(\lambda) = \lambda - \frac{\lambda^2}{4}.$$

Maximize: $q'(\lambda) = 1 - \lambda/2 = 0 \implies \lambda = 2$. Dual value $q(2) = 1$. Primal optimum: at $x = 1$, value 1. Dual matches primal $\implies$ strong duality.

Example B: Equality Constraint

$$\min_x (x - 3)^2 \quad \text{s.t.} \quad x = 1.$$

Lagrangian: $L(x, \nu) = (x - 3)^2 + \nu(x - 1)$. Stationarity: $2(x - 3) + \nu = 0 \implies x = 3 - \nu/2$. Constraint $x = 1 \implies \nu = 4$. Primal optimum: $f(1) = 4$. Dual evaluation also gives 4.

Example B:

Primal Problem

$$\min_{x \in \mathbb{R}^2} x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

Primal Solution

Since x_1 is cheaper than x_2 , put all weight on x_1 : $x_1 = 1, x_2 = 0 \implies \text{cost} = 1$. This is optimal.

Dual Problem

Write in standard form with $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, $b = (1, 0, 0)^T$, $c = (1, 2)^T$. Dual:

$$\max_{y \geq 0} b^T y \quad \text{s.t.} \quad A^T y \leq c.$$

That is

$$\max_{y \geq 0} y_1 \quad \text{s.t.} \quad y_1 + y_2 \leq 1, \quad y_1 + y_3 \leq 2.$$

Dual Solution

Choose $y_1 = 1, y_2 = 0, y_3 = 0 \implies \text{value} = 1$. Dual value = primal value $\implies$ strong duality holds.

5. Quadratic Programming (QP) Example

Example C:

Primal Problem

$$\min_{x \in \mathbb{R}^2} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad x_1 + x_2 \geq 1, \quad x_1 \geq 0, \quad x_2 \geq 0,$$

with $Q = 2I_2$, $c = (-2, -5)^T$.

Primal Solution

Stationarity of unconstrained problem: $Qx + c = 0 \implies x = (1, 2.5)$. It satisfies all constraints, so it is primal optimum. Objective value:

$$f(x) = (1^2 + 2.5^2) - (2 + 12.5) = 7.25 - 14.5 = -7.25.$$

Dual Problem

With constraint matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, $b = (1, 0, 0)^T$, multipliers $\lambda \geq 0$:

$$q(\lambda) = -\frac{1}{4} \|(A^T \lambda - c)\|^2 + b^T \lambda.$$

Explicitly: $A^T \lambda = [\lambda_1 + \lambda_2, \lambda_1 + \lambda_3]^T$.

$$q(\lambda) = -\frac{1}{4} ((\lambda_1 + \lambda_2 + 2)^2 + (\lambda_1 + \lambda_3 + 5)^2) + \lambda_1.$$

Dual Solution

At $\lambda = 0$: $q(0) = -7.25$, equal to primal optimum. So $\lambda^* = 0$ is dual-optimal. Strong duality holds.

Example D:

Minimize

$$\frac{1}{2} x^\top Q x + c^\top x, \quad Q = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \succ 0, \quad c = \begin{bmatrix} -2 \\ -6 \end{bmatrix},$$

subject to one inequality and one equality:

$$Ax \geq b, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b = 1, \quad Ex = d, \quad E = \begin{bmatrix} 1 & -1 \end{bmatrix}, \quad d = 0.$$

Equivalently: $x_1 + x_2 \geq 1$ and $x_1 - x_2 = 0$.

Primal Solution

From the equality $x_1 = x_2 = t$. The inequality gives $2t \geq 1 \Rightarrow t \geq 0.5$. The objective as a function of t :

$$f(t) = \frac{1}{2} \begin{bmatrix} t & t \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} t \\ t \end{bmatrix} + \begin{bmatrix} -2 & -6 \end{bmatrix} \begin{bmatrix} t \\ t \end{bmatrix} = \frac{1}{2} (2t^2 + 4t^2) - 8t = 3t^2 - 8t.$$

Unconstrained minimizer: $f'(t) = 6t - 8 = 0 \Rightarrow t = \frac{4}{3}$, which satisfies $t \geq 0.5$.

Primal optimum: $x^* = (\frac{4}{3}, \frac{4}{3})$.

Objective value:

$$f(x^*) = 3\left(\frac{4}{3}\right)^2 - 8\left(\frac{4}{3}\right) = \frac{16}{3} - \frac{32}{3} = -\frac{16}{3}.$$

KKT Multipliers and Checks

Lagrangian

$$L(x, \lambda, \nu) = \frac{1}{2}x^\top Qx + c^\top x + \lambda(b - Ax) + \nu(d - Ex),$$

with $\lambda \geq 0$ (inequality), $\nu \in \mathbb{R}$ (equality).

Stationarity: $Qx + c - A^\top \lambda - E^\top \nu = 0$.

At x^* ,

$$Qx^* + c = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 4/3 \\ 4/3 \end{bmatrix} + \begin{bmatrix} -2 \\ -6 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -2/3 \end{bmatrix}.$$

Complementary slackness: $Ax^* - b = 8/3 - 1 = 5/3 > 0$ so the inequality is slack $\Rightarrow \lambda^* = 0$.

Stationarity reduces to

$$\begin{bmatrix} 2/3 \\ -2/3 \end{bmatrix} - E^\top \nu^* = \begin{bmatrix} 2/3 - \nu^* \\ -2/3 + \nu^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \nu^* = \frac{2}{3}.$$

Feasibility holds; complementary slackness $\lambda^*(Ax^* - b) = 0$ holds. KKT satisfied.

Dual Value (Matches Primal)

For QP with equality $Ex = d$ and inequality $Ax \geq b$, the dual is

$$\max_{\lambda \geq 0, \nu \text{ free}} q(\lambda, \nu) = -\frac{1}{2}(A^\top \lambda + E^\top \nu - c)^\top Q^{-1}(A^\top \lambda + E^\top \nu - c) + b^\top \lambda + d^\top \nu,$$

with $Q^{-1} = \text{diag}(1/2, 1/4)$.

At $(\lambda^*, \nu^*) = (0, \frac{2}{3})$,

$$s := A^\top \lambda^* + E^\top \nu^* - c = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \nu^* \\ -\nu^* \end{bmatrix} - \begin{bmatrix} -2 \\ -6 \end{bmatrix} = \begin{bmatrix} \frac{8}{3} \\ \frac{16}{3} \end{bmatrix},$$

$$Q^{-1}s = \begin{bmatrix} (1/2) \cdot 8/3 \\ (1/4) \cdot 16/3 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 4/3 \end{bmatrix}, \quad s^\top Q^{-1}s = \frac{8}{3} \cdot \frac{4}{3} + \frac{16}{3} \cdot \frac{4}{3} = \frac{32}{3}.$$

Hence

$$q(\lambda^*, \nu^*) = -\frac{1}{2} \cdot \frac{32}{3} + b^\top \lambda^* + d^\top \nu^* = -\frac{16}{3} + 0 + 0 = -\frac{16}{3},$$

which equals the primal optimum.

Summary

Primal optimum $x^* = (\frac{4}{3}, \frac{4}{3})$ with value $-\frac{16}{3}$; multipliers $\lambda^* = 0$, $\nu^* = \frac{2}{3}$; dual value $-\frac{16}{3}$ (strong duality).

Duality Theory

Consider the optimization problem

$$\begin{aligned} \text{Primal (P): } \quad & \min_{x \in \mathbb{R}^n} f(x) \\ \text{s.t. } & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned}$$

where f and g_i are convex functions and h_j are affine functions.

The associated Lagrangian is

$$L(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x),$$

with $\lambda \geq 0$, ν free.

The *dual function* is

$$q(\lambda, \nu) = \inf_x L(x, \lambda, \nu).$$

The *dual problem* is

$$\text{maximize } q(\lambda, \nu) \quad \text{s.t. } \lambda \geq 0.$$

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Weak Duality Theorem

Theorem (Weak Duality). Let x be any feasible solution of the primal problem and (λ, ν) any feasible solution of the dual problem. Then

$$q(\lambda, \nu) \leq f(x).$$

Equivalently, the optimal value of the dual problem is always less than or equal to the optimal value of the primal problem:

$$d^* \leq p^*.$$

Proof (sketch). For any feasible x , $\lambda \geq 0$, and ν ,

$$L(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x).$$

Since $g_i(x) \leq 0$ and $\lambda_i \geq 0$, and $h_j(x) = 0$, we have $L(x, \lambda, \nu) \leq f(x)$. Taking $\inf_x L(x, \lambda, \nu) = q(\lambda, \nu) \leq L(x, \lambda, \nu) \leq f(x)$. $\square$

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Strong Duality Theorem

Theorem (Strong Duality). Suppose the primal problem is convex (i.e., f and g_i are convex, h_j affine) and satisfies a regularity condition such as *Slater's condition* (existence of a strictly feasible point x with $g_i(x) < 0$, $h_j(x) = 0$). Then

$$d^* = p^*.$$

That is, the optimal values of the primal and dual problems coincide.

Remarks.

- Weak duality holds universally, even for non-convex problems.
- Strong duality requires convexity and a constraint qualification (e.g., Slater's condition).
- When strong duality holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality.

6. Key Takeaways

- The **Lagrangian** folds constraints into the objective using multipliers.
- The **dual function** comes from minimizing the Lagrangian over primal variables.
- The **dual problem** is maximizing this function over the multipliers.
- Numerical examples show how dual values match primal values (strong duality) for convex problems.
- LP and QP duals are central in optimization theory and practical algorithms.